



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/31

Paper 3 Pure Mathematics 3 (P3)

October/November 2009

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of **3** printed pages and **1** blank page.



UNIVERSITY of CAMBRIDGE
International Examinations

- 1 Solve the inequality $2 - 3x < |x - 3|$. [4]

- 2 Solve the equation $3^{x+2} = 3^x + 3^2$, giving your answer correct to 3 significant figures. [4]

- 3 The sequence of values given by the iterative formula

$$x_{n+1} = \frac{3x_n}{4} + \frac{15}{x_n^3},$$

with initial value $x_1 = 3$, converges to α .

- (i) Use this iterative formula to find α correct to 2 decimal places, giving the result of each iteration to 4 decimal places. [3]

- (ii) State an equation satisfied by α and hence find the exact value of α . [2]

- 4 A curve has equation $y = e^{-3x} \tan x$. Find the x -coordinates of the stationary points on the curve in the interval $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$. Give your answers correct to 3 decimal places. [6]

- 5 (i) Prove the identity $\cos 4\theta - 4\cos 2\theta + 3 \equiv 8\sin^4 \theta$. [4]

- (ii) Using this result find, in simplified form, the exact value of

$$\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \sin^4 \theta \, d\theta. \quad [4]$$

- 6 With respect to the origin O , the points A , B and C have position vectors given by

$$\overrightarrow{OA} = \mathbf{i} - \mathbf{k}, \quad \overrightarrow{OB} = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k} \quad \text{and} \quad \overrightarrow{OC} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}.$$

The mid-point of AB is M . The point N lies on AC between A and C and is such that $AN = 2NC$.

- (i) Find a vector equation of the line MN . [4]

- (ii) It is given that MN intersects BC at the point P . Find the position vector of P . [4]

- 7 The complex number $-2 + i$ is denoted by u .

- (i) Given that u is a root of the equation $x^3 - 11x - k = 0$, where k is real, find the value of k . [3]

- (ii) Write down the other complex root of this equation. [1]

- (iii) Find the modulus and argument of u . [2]

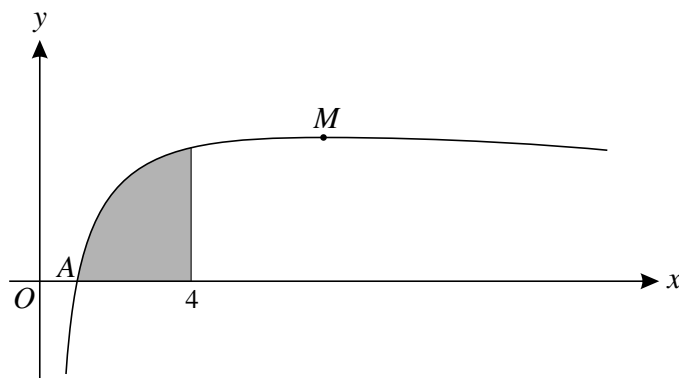
- (iv) Sketch an Argand diagram showing the point representing u . Shade the region whose points represent the complex numbers z satisfying both the inequalities

$$|z| < |z - 2| \quad \text{and} \quad 0 < \arg(z - u) < \frac{1}{4}\pi. \quad [4]$$

- 8 (i) Express $\frac{5x+3}{(x+1)^2(3x+2)}$ in partial fractions. [5]

- (ii) Hence obtain the expansion of $\frac{5x+3}{(x+1)^2(3x+2)}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients. [5]

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The diagram shows the curve $y = \frac{\ln x}{\sqrt{x}}$ and its maximum point M . The curve cuts the x -axis at the point A .

- (i) State the coordinates of A . [1]
- (ii) Find the exact value of the x -coordinate of M . [4]
- (iii) Using integration by parts, show that the area of the shaded region bounded by the curve, the x -axis and the line $x = 4$ is equal to $8 \ln 2 - 4$. [5]
- 10 In a model of the expansion of a sphere of radius r cm, it is assumed that, at time t seconds after the start, the rate of increase of the surface area of the sphere is proportional to its volume. When $t = 0$, $r = 5$ and $\frac{dr}{dt} = 2$.

- (i) Show that r satisfies the differential equation

$$\frac{dr}{dt} = 0.08r^2. \quad [4]$$

[The surface area A and volume V of a sphere of radius r are given by the formulae $A = 4\pi r^2$, $V = \frac{4}{3}\pi r^3$.]

- (ii) Solve this differential equation, obtaining an expression for r in terms of t . [5]
- (iii) Deduce from your answer to part (ii) the set of values that t can take, according to this model. [1]

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